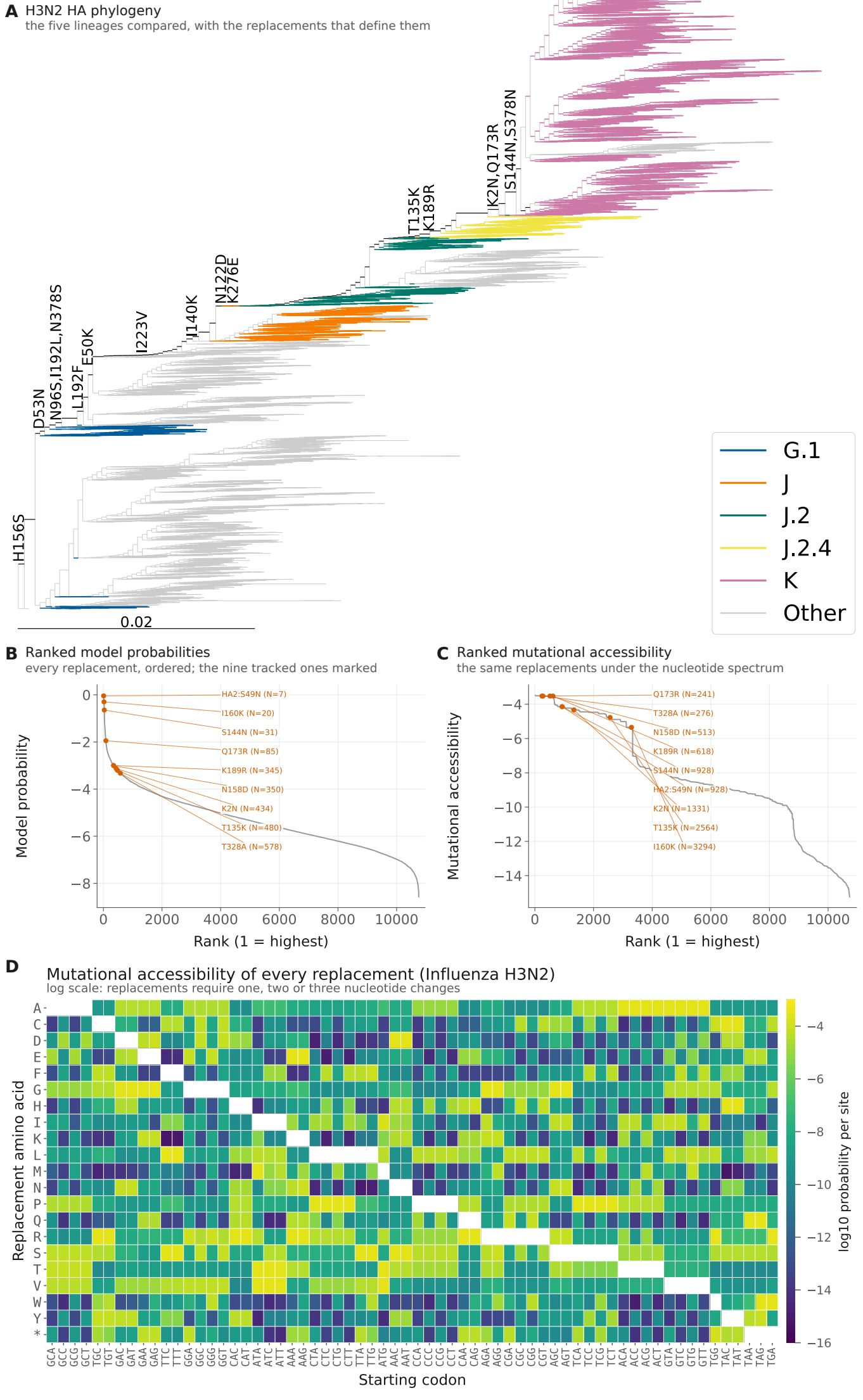

A
H3N2 HA phylogeny
the five lineages compared, with the replacements that define them
H156S
D53N
N96S,I192L,N378S
L192F
E50K
I223V
I140K
N122D
K276E
T135K
K189R
K2N,Q173R
S144N,S378N
0.02
G.1
J
J.2
J.2.4
K
Other
B
Ranked model probabilities
every replacement, ordered; the nine tracked ones marked
Model probability
HA2:S49N (N=7)
I160K (N=20)
S144N (N=31)
Q173R (N=85)
K189R (N=345)
N158D (N=350)
K2N (N=434)
T135K (N=480)
T328A (N=578)
Rank (1 = highest)
C
Ranked mutational accessibility
the same replacements under the nucleotide spectrum
Mutational accessibility
Q173R (N=241)
T328A (N=276)
N158D (N=513)
K189R (N=618)
S144N (N=928)
HA2:S49N (N=928)
K2N (N=1331)
T135K (N=2564)
I160K (N=3294)
Rank (1 = highest)
D
Mutational accessibility of every replacement (Influenza H3N2)
log scale: replacements require one, two or three nucleotide changes
Replacement amino acid
Starting codon
log10 probability per site